CLASS 9 · JEE ADVANCED · TOP-10 QUESTION IMPACT REPORT

Pragya G9 (EXI) Achievers

Part Test-1 · Paper 2 · which questions cost the top 10 the most

Exam date 09 Oct 2026
Campus HYD/Nallakunta DD Colony
Students analysed top 10 by program rank

In total, the top 10 lost **≈ 656 marks** – **≈ 103** from Maths, **≈ 297** from Physics and **≈ 256** from Chemistry.



Marks lost = marks not earned on wrong or skipped questions plus negative marks. Approximate (≈) because multiple-correct questions give partial marks.

16 of 50
high-impact questions – lost by 5+ of the top 10

45%
of all lost marks came from Physics

Q51 10/10
lost by every student

106 / 180
top-10 average score (range 84–139)

1 Where the damage happened

Marks the top 10 gave away in each subject, split into marks lost on **wrong answers** (the question's marks plus the negative mark) and marks left on **skipped** questions. Advanced figures are approximate (≈) – see note at the end.

SUBJECT	SPLIT ■ wrong ■ skipped	MARKS LOST	SHARE	ACCURACY*	HIGH-IMPACT
Maths	≈ 85 on 24 wrong answers (incl. 3 negative) · 18 on 5 skipped	**≈ 103** ≈ 10 per student	16%	85%	**2** of 17 Qs
Physics	≈ 225 on 59 wrong answers (incl. 24 negative) · 72 on 21 skipped	**≈ 297** ≈ 30 per student	45%	60%	**8** of 17 Qs
Chemistry	≈ 245 on 67 wrong answers (incl. 14 negative) · 11 on 3 skipped	**≈ 256** ≈ 26 per student	39%	57%	**6** of 16 Qs

* Accuracy = right answers ÷ questions attempted.

BY QUESTION TYPE	MATHS	PHYSICS	CHEMISTRY
Single correct marking +3 / −1	Q1-4 **8%** wrong · **2%** skipped	Q18-21 **45%** wrong · **8%** skipped	Q35-38 **20%** wrong · **0%** skipped
Multiple correct marking +4 / −2	Q5-7 **0%** wrong · **3%** skipped	Q22-24 **10%** wrong · **10%** skipped	Q39-41 **10%** wrong · **0%** skipped
Integer marking +4 / 0	Q8-13 **17%** wrong · **3%** skipped	Q25-30 **35%** wrong · **10%** skipped	Q42-47 **54%** wrong · **4%** skipped
Paragraph marking +3 / 0	Q14-17 **28%** wrong · **2%** skipped	Q31-34 **42%** wrong · **22%** skipped	Q48-51 **72%** wrong · **2%** skipped

Key takeaways

1. **Physics** did the most damage: **45%** of all marks lost by the top 10 (≈ 297 of ≈ 656). 8 of its 17 questions were lost by 5 or more students.
2. **Q51** was lost by all 10 students, and **4 more** by 9 of 10 (Q18, Q27, Q33, Q46).
3. Biggest chapter-level loss: **Physics · Friction** – ≈ 124 marks across 5 questions (4 high-impact).
4. **Paragraph** questions were the weakest type in the paper: **57%** of answers wrong or blank, against 36% across all questions.
5. Skipping is concentrated in **Physics · Paragraph**: 9 of 40 answers (22%) were left blank.
6. Negative marking alone cost the top 10 **41 marks** (about 4 per student).
7. Strong ground: all 10 answered **8 questions** correctly (Q2, Q5, Q6, Q9, Q15, Q37, Q39, Q40).

Answered wrong by 5+ students

Students attempted these and still got them wrong – points to a misconception or a trap worth re-teaching.

Q51 10 wrong · **Q18** 8 wrong · **Q46** 8 wrong · **Q48** 8 wrong · **Q20** 7 wrong · **Q47** 7 wrong · **Q50** 7 wrong · **Q17** 6 wrong · **Q33** 6 wrong · **Q8** 5 wrong · **Q27** 5 wrong · **Q29** 5 wrong · **Q30** 5 wrong · **Q32** 5 wrong · **Q43** 5 wrong

Left blank by 5+ students

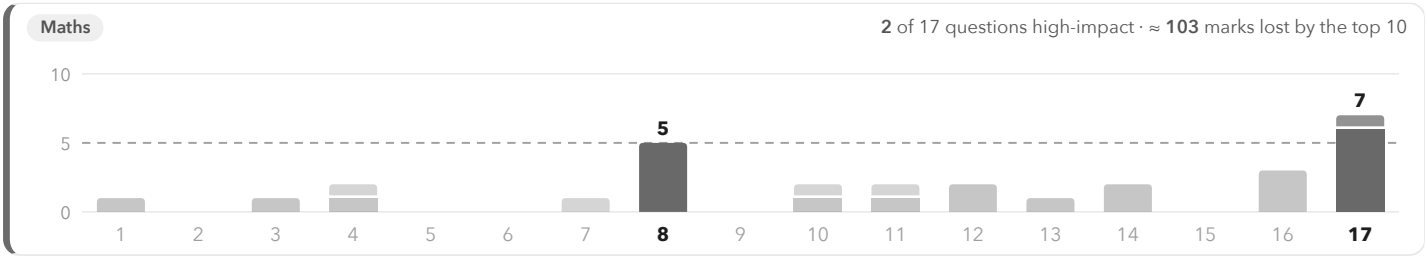
Most students did not attempt these – points to an unfamiliar topic, low confidence, or running out of time.

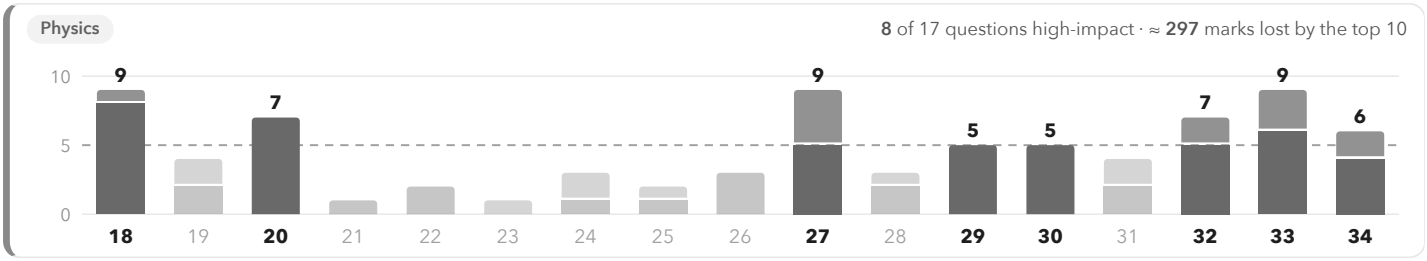
None

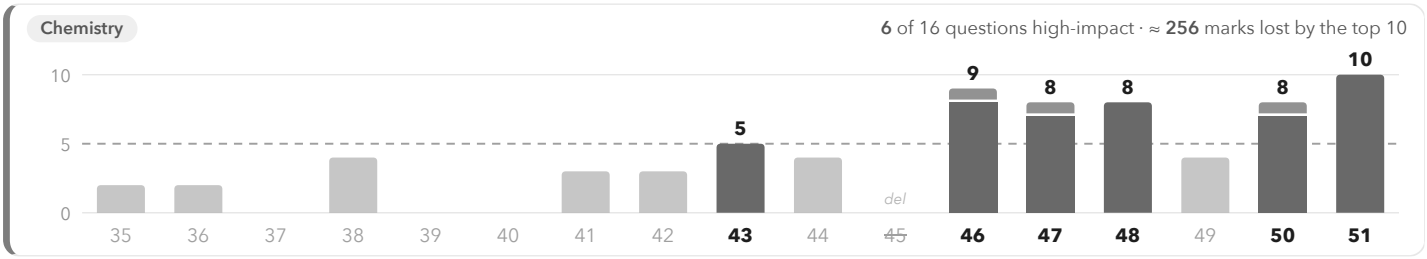
2 Question-by-question impact

How many of the top 10 lost each question. Questions at or above the dashed line (lost by 5+) are **high-impact** and shown in full colour with their count.

■ Wrong answer ■ Skipped ---- High-impact line (5 of 10) ■ Below the line (faded) "del" = deleted / bonus question







3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

Marks lost adds up what all 10 students gave away on a chapter's questions: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Physics · Friction** has 5 questions, and the top 10 lost ≈ **124** marks on them — about **12 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 6.8 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics ≈ 297 marks lost · 45% of all lost marks · 8 high-impact questions					
Friction	5	**Q18, Q27, Q30, Q32**	**≈ 124**	≈ 12 marks	6.8 of 10
Work, Energy & Power	4	**Q20, Q33, Q34**	**≈ 77**	≈ 8 marks	5.8 of 10
Circular Motion	4	**Q29**	**≈ 60**	≈ 6 marks	3.8 of 10
Laws of Motion	3	None reached 5 · most missed Q22, Q25 (2/10)	**≈ 24**	≈ 2 marks	1.7 of 10
Momentum & Collisions	1	None reached 5 · most missed Q28 (3/10)	**≈ 12**	≈ 1 marks	3.0 of 10
2. Chemistry ≈ 256 marks lost · 39% of all lost marks · 6 high-impact questions					
Hydrocarbons	5	**Q43, Q47, Q50, Q51**	**≈ 106**	≈ 11 marks	6.2 of 10
General Organic Chemistry	9	**Q48**	**≈ 96**	≈ 10 marks	3.0 of 10
Isomerism	2	**Q46**	**≈ 54**	≈ 5 marks	6.0 of 10
3. Maths ≈ 103 marks lost · 16% of all lost marks · 2 high-impact questions					
Functions	6	**Q8**	**≈ 47**	≈ 5 marks	2.0 of 10
Trigonometric Equations	5	**Q17**	**≈ 42**	≈ 4 marks	2.6 of 10
Inverse Trigonometric Functions	6	None reached 5 · most missed Q14 (2/10)	**≈ 14**	≈ 1 marks	0.7 of 10

4 High-impact questions at a glance

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

Marks lost = marks the top 10 together gave away on that question: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Q51** — 10 wrong × 3 = 30 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

Critical lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths

2 of 17 questions high-impact · ≈ **41** marks lost on them by the top 10

Q	CHAPTER — WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q17	**Trigonometric Equations** — Sum of solutions of $\cos(n_1\sqrt{(x-100)})\cdot\cos(n_1\sqrt{x}) = 1$	Paragraph · +3 / 0	**7/10**	6	1	**≈ 21**
Q8	**Functions** — Period from two lines of symmetry	Integer · +4 / 0	**5/10**	5	0	**≈ 20**

Physics

8 of 17 questions high-impact · ≈ **205** marks lost on them by the top 10

Q	CHAPTER — WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q18	**Friction** — Static vs kinetic friction under a time-varying force	Single · +3 / −1	**9/10**	8	1	**≈ 35**
Q33	**Work, Energy & Power** — Work-energy theorem from a-t and U-x graphs	Paragraph · +3 / 0	**9/10**	6	3	**≈ 27**
Q27	**Friction** — Net work done by friction between sliding blocks	Integer · +4 / 0	**9/10**	5	4	**≈ 36**
Q20	**Work, Energy & Power** — Maximum spring extension — energy method	Single · +3 / −1	**7/10**	7	0	**≈ 28**
Q32	**Friction** — Total force exerted by a block on an accelerating cube	Paragraph · +3 / 0	**7/10**	5	2	**≈ 21**
Q34	**Work, Energy & Power** — Work by conservative forces from the U-x graph	Paragraph · +3 / 0	**6/10**	4	2	**≈ 18**
Q29	**Circular Motion** — Net acceleration in a vertical circle at 37°	Integer · +4 / 0	**5/10**	5	0	**≈ 20**
Q30	**Friction** — Work by friction seen from an accelerating observer	Integer · +4 / 0	**5/10**	5	0	**≈ 20**

Chemistry

6 of 16 questions high-impact · ≈ **166** marks lost on them by the top 10

Q	CHAPTER — WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q51	**Hydrocarbons** — Chiral products of free-radical monobromination	Paragraph · +3 / 0	**10/10**	10	0	**≈ 30**
Q46	**Isomerism** — Count of optically active compounds	Integer · +4 / 0	**9/10**	8	1	**≈ 36**
Q48	**General Organic Chemistry** — Counting electrophiles	Paragraph · +3 / 0	**8/10**	8	0	**≈ 24**
Q47	**Hydrocarbons** — Dichlorination isomers of C_5H_{10} (VD 35)	Integer · +4 / 0	**8/10**	7	1	**≈ 32**
Q50	**Hydrocarbons** — Preparation of alkanes — count correct statements	Paragraph · +3 / 0	**8/10**	7	1	**≈ 24**
Q43	**Hydrocarbons** — Alkane from O_2 needed; count secondary carbons	Integer · +4 / 0	**5/10**	5	0	**≈ 20**

5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 180, **Avg** = average of the 6 weekly Advanced tests (WTA), out of 180. ✗ = wrong, – = skipped, ✓ = correct.

■ ✗ Wrong ■ – Skipped ■ ✓ Correct

	Rank & score				Maths		Physics						Chemistry						
STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	8	17	18	20	27	29	30	32	33	34	43	46	47	48	50	51	Lost
Prit Raj Kushwaha	1	**4**	139	118.3*	✓	✓	✗	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	**7**
Puneeth K. Jureddi	2	**7**	136	125.8*	✓	✓	✓	✗	–	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✗	**5**
Geetansh P. R. Vengala	3	**8**	135	101.3*	✓	✓	✗	✓	✗	✗	✓	✓	✗	✓	✗	✗	✗	✗	–	✗	**10**
Ramini Havisha	4	**28**	110	101.4	✗	✗	✗	✗	–	✓	✓	–	✗	✓	✓	–	–	✗	✓	✗	**11**
Nandimandalam L. Reddy	6	**63**	95	108.2*	✗	✗	–	✗	✗	✓	✗	✗	–	✗	✗	✗	✓	✗	✗	✗	**14**
Ankam Advaith	9	**71**	92	83.7*	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	**15**
Bashaboina Sai Saketh	10	**72**	92	63.8	✓	–	✗	✗	–	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	**13**
Rudroju Rohan Chary	11	**87**	91	72.2*	✗	✗	✗	✗	✓	✗	✗	–	–	–	✓	✗	✗	✓	✗	✗	**13**
Bandhav Praputh	14	**101**	85	81.3*	✓	✗	✗	✓	✗	✗	✗	✗	–	–	✗	✗	✗	✗	✗	✗	**14**
Pamunugundla Smaran	15	**108**	84	80.4*	✗	✗	✗	✗	–	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	**15**
Students who lost it					5	7	9	7	9	5	5	7	9	6	5	9	8	8	8	10	

6 Subject scores of the top 10

Estimated subject scores (≈) from right / wrong / skipped answers; partial credit on multiple-correct questions is not in the source data, so these can run a few marks above the official total. The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Chemistry 56 = 60 minus Q45 (4).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	Maths score · right / wrong / skipped	Physics score · right / wrong / skipped	Chemistry score · right / wrong / skipped	WEAKEST
Prit Raj Kushwaha	1	**4**	**139**	118.3*	≈ **57**/60 16 / 1 / 0	≈ **52**/60 15 / 2 / 0	≈ **32**/56 9 / 7 / 0	Chemistry
Puneeth K. Jureddi	2	**7**	**136**	125.8*	≈ **60**/60 17 / 0 / 0	≈ **40**/60 12 / 4 / 1	≈ **41**/56 12 / 4 / 0	Physics
Geetansh P. R. Vengala	3	**8**	**135**	101.3*	≈ **57**/60 16 / 1 / 0	≈ **45**/60 13 / 4 / 0	≈ **35**/56 10 / 5 / 1	Chemistry
Ramini Havisha	4	**28**	**110**	101.4	≈ **49**/60 14 / 2 / 1	≈ **31**/60 9 / 3 / 5	≈ **35**/56 10 / 4 / 2	Physics
Nandimandalam L. Reddy	6	**63**	**95**	108.2*	≈ **46**/60 13 / 4 / 0	≈ **25**/60 7 / 7 / 3	≈ **35**/56 10 / 6 / 0	Physics
Ankam Advaith	9	**71**	**92**	83.7*	≈ **49**/60 14 / 3 / 0	≈ **27**/60 8 / 9 / 0	≈ **21**/56 7 / 9 / 0	Chemistry
Bashaboina Sai Saketh	10	**72**	**92**	63.8	≈ **39**/60 11 / 5 / 1	≈ **34**/60 10 / 6 / 1	≈ **21**/56 7 / 9 / 0	Chemistry
Rudroju Rohan Chary	11	**87**	**91**	72.2*	≈ **42**/60 12 / 4 / 1	≈ **17**/60 5 / 5 / 7	≈ **38**/56 11 / 5 / 0	Physics
Bandhav Praputh	14	**101**	**85**	81.3*	≈ **53**/60 15 / 2 / 0	≈ **16**/60 6 / 8 / 3	≈ **18**/56 6 / 10 / 0	Physics
Pamunugundla Smaran	15	**108**	**84**	80.4*	≈ **45**/60 13 / 2 / 2	≈ **16**/60 5 / 11 / 1	≈ **28**/56 8 / 8 / 0	Physics

Average. "Avg" = the student's average over the 6 weekly Advanced tests (WTA) before this Part Test, out of 180 (those tests are marked out of 240; averages are scaled to 180 to match this paper — × 180/240, percentages unchanged), from the campus Mains & Advance averages sheet. * Sat fewer weekly tests (tests sat / conducted): Prit Raj Kushwaha (5/6), Puneeth Karthikeya Jureddi (4/6), Geetansh Praveen Reddy Vengala (5/6), Nandimandalam Loukhya Reddy (5/6), Ankam Advaith (4/6), Rudroju Rohan Chary (5/6), Bandhav Praputh (5/6), Pamunugundla Smaran (5/6).

Ranks. Program rank = rank within Pragya G9 (EXI) Achievers across all campuses (from the final marks sheet; the top 10 are picked by this rank). **AIR** = All-India rank across the whole Class 9 programme (all batches, all campuses), from the institute's All India Marks Analysis sheet.

How to read this report. "Lost" means the student did not get the question right — either answered it wrong or left it blank. A question is **high-impact** when 5 or more of the top 10 lost it. **"Marks lost"** = marks the top 10 together gave away: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. "Per student" divides that by 10. **Deleted question:** Q45 has no right / wrong / skipped entry for any student in the final marks sheet (deleted / bonus question), so it is left out of all counts and shown as "del" in the charts. Chapter tags were assigned by reading each question; please correct any that your department classifies differently.

Note on Advanced marks. The paper uses partial marking for multiple-correct questions, but the source data only says right / wrong / skipped. Marks-lost figures use the paper's marking scheme (shown on every question) and are approximate; the counts of students who lost each question are exact.

7 Maths – high-impact questions

2 questions, worst first
cropped from the question paper

Q17 Trigonometric Equations – Sum of solutions of cos(π√(x−100))·cos(π√x) = 1

Maths · Paragraph · marking +3 / 0 · marks lost by the top 10: 6 wrong × 3 + 1 skipped × 3 = 21

Lost by 7/10
6 wrong 1 skipped

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2 - m^2 = k$ then we use $(n+m)(n-m) = k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n,m).

Eg: $n^2 - m^2 = 6 \Rightarrow (n+m)(n-m) = (\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

17. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-100}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is .....

Wrong: Ramini Havisha, Nandimandalam Loukhya Reddy, Ankam Advaith, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

Skipped: Bashaboina Sai Saketh

Q8 Functions – Period from two lines of symmetry

Maths · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 = 20

Lost by 5/10
5 wrong 0 skipped

8. Let $a \in R$. If $f(a-x) = f(a+x) \forall x \in R$, then $x = a$ is a line of symmetry in the graph of $y = f(x)$. Consider a non-constant function $g(x)$ such that $g(1+x) = g(1-x) \forall x \in R$ and $g(2026+x) = g(2026-x) \forall x \in R$. Then all such $g(x)$ must be periodic. The smallest positive number that could be a period for all such $g(x)$ is .....

Wrong: Ramini Havisha, Nandimandalam Loukhya Reddy, Ankam Advaith, Rudroju Rohan Chary, Pamunugundla Smaran

Skipped: –

Physics – high-impact questions

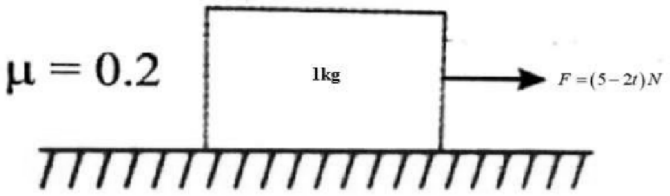
8 questions, worst first
cropped from the question paper

Q18 Friction – Static vs kinetic friction under a time-varying force

Physics · Single correct · marking +3 / −1 · marks lost by the top 10: 8 wrong × 4 + 1 skipped × 3 = 35

Lost by 9/10
8 wrong · 1 skipped

18. The force acting on the block of mass 1 kg is given by $F = 5 - 2t$. The frictional force acting on the block after time $t = 2\text{sec}$ will be $\left(\mu = 0.2, g = 10 m/s^2\right)$



A) 2N B) 3N C) 1N D) zero

Wrong: Prit Raj Kushwaha, Geetansh Praveen Reddy Vengala, Ramini Havisha, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

Skipped: Nandimandalam Loukhya Reddy

Q33 Work, Energy & Power – Work-energy theorem from a-t and U-x graphs

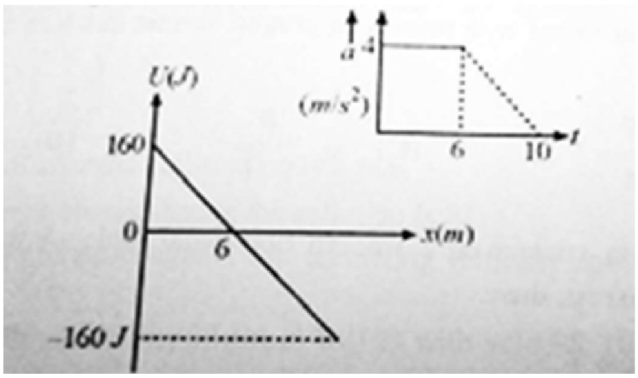
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 6 wrong × 3 + 3 skipped × 3 = 27

Lost by 9/10
6 wrong · 3 skipped

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



33. Work done by all forces _____ J

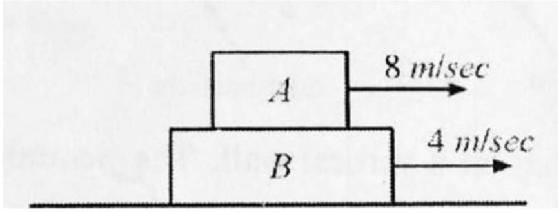
Wrong: Puneeth Karthikeya Jureddi, Geetansh Praveen Reddy Vengala, Ramini Havisha, Ankam Advaith, Bashaboina Sai Saketh, Pamunugundla Smaran

Skipped: Nandimandalam Loukhya Reddy, Rudroju Rohan Chary, Bandhav Praputh

Q27 **Friction** – Net work done by friction between sliding blocks
Physics · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 + 4 skipped × 4 = 36

Lost by 9/10
5 wrong · 4 skipped

27. Block A of mass 1kg is placed on rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given velocities as shown. If net workdone by frictional force is "-n" J. value of n is ……..



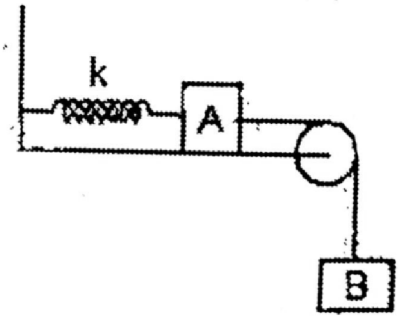
Wrong: Prit Raj Kushwaha, Geetansh Praveen Reddy Vengala, Nandimandalam Loukhya Reddy, Ankam Advaith, Bandhav Praputh

Skipped: Puneeth Karthikeya Jureddi, Ramini Havisha, Bashaboina Sai Saketh, Pamunugundla Smaran

Q20 **Work, Energy & Power** – Maximum spring extension – energy method
Physics · Single correct · marking +3 / −1 · marks lost by the top 10: 7 wrong × 4 = 28

Lost by 7/10
7 wrong · 0 skipped

20. All the surfaces are smooth system is released from rest with spring unstretched. Then maximum extension in spring will be $(m_A = m; m_B = 2m)$



A) $\frac{mg}{K}$ B) $\frac{2mg}{K}$ C) $\frac{3mg}{K}$ D) $\frac{4mg}{K}$

Wrong: Puneeth Karthikeya Jureddi, Ramini Havisha, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Pamunugundla Smaran

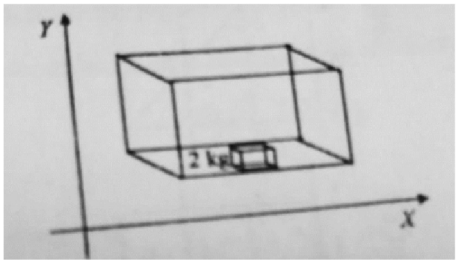
Skipped: –

Q32 **Friction** – Total force exerted by a block on an accelerating cube
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 + 2 skipped × 3 = 21

Lost by 7/10
5 wrong 2 skipped

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V} = \left(5t\hat{i} + 2t\hat{j}\right) m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu = 0.5$ between block and surface of cube. $\left(g = 10 m/s^2\right)$



32. Total force exerted by block on cube is _____ N

Wrong: Puneeth Karthikeya Jureddi, Nandimandalam Loukhya Reddy, Ankam Advaith, Bandhav Praputh, Pamunugundla Smaran

Skipped: Ramini Havisha, Rudroju Rohan Chary

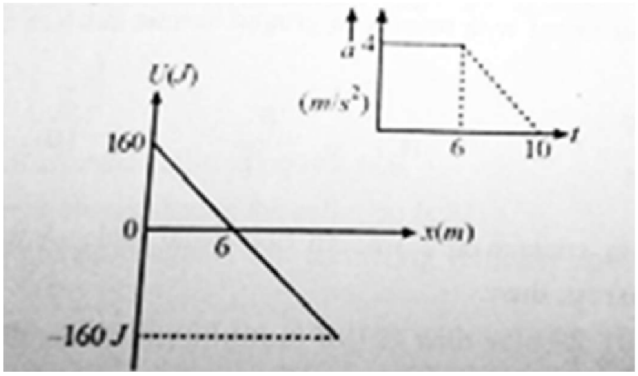
Q34 **Work, Energy & Power** – Work by conservative forces from the U-x graph
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 4 wrong × 3 + 2 skipped × 3 = 18

Lost by 6/10
4 wrong 2 skipped

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



34. Work done by conservative forces _____ J

Wrong: Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Pamunugundla Smaran

Skipped: Rudroju Rohan Chary, Bandhav Praputh

Q29 **Circular Motion** – Net acceleration in a vertical circle at 37°
Physics · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 = 20

Lost by 5/10
5 wrong 0 skipped

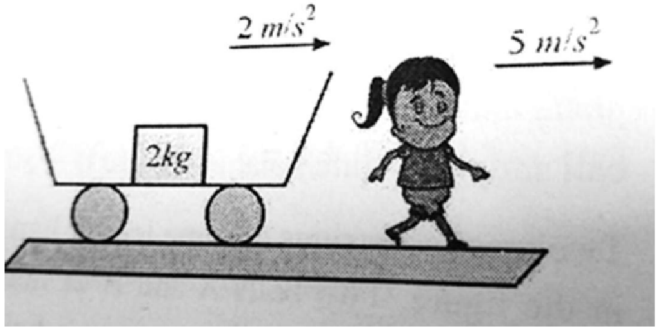
29. A small ball of mass m is attached to the end of string of length $l = 1m$ whose other end is fixed. From its lowest position, the ball is given a kinetic energy $\frac{mgl}{5}$. Net acceleration $\left(in\ m/s^2\right)$ of ball at the instant when string makes an angle $\theta = 37^o$ with vertical is ..... $\left(g = 10m/s^2\right)$

Wrong: Geetansh Praveen Reddy Vengala, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran **Skipped:** –

Q30 **Friction** – Work by friction seen from an accelerating observer
Physics · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 = 20

Lost by 5/10
5 wrong 0 skipped

30. An observer and a vehicle, both starts moving together from rest with acceleration $5m/s^2$ and $2m/s^2$ respectively. There is a 2 kg block on the floor of the vehicle and $\mu = 0.3$ between their surfaces. The work done by the frictional force on 2kg block as observed by running observer , during first 2seconds of motion is $\underline{-x}$ (J) $\left(g = 10m/s^2\right)$ find 'x'



Wrong: Nandimandalam Loukhya Reddy, Ankam Advaith, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran **Skipped:** –

Q51 Hydrocarbons – Chiral products of free-radical monobromination

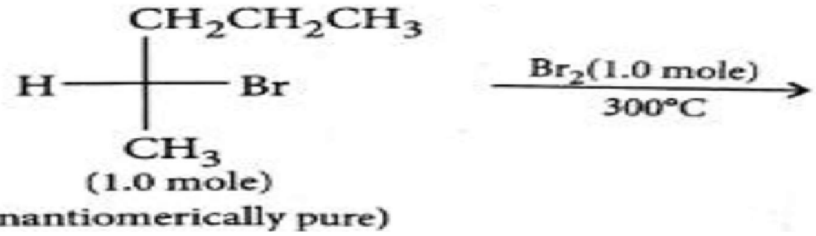
Chemistry · Paragraph · marking +3 / 0 · marks lost by the top 10: 10 wrong × 3 = 30

Lost by 10/10
10 wrong · 0 skipped

PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

51. In the following monobromination reaction, the number of possible chiral products is



Wrong: Prit Raj Kushwaha, Puneeth Karthikeya Jureddi, Geetansh Praveen Reddy Vengala, Ramini Havisha, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

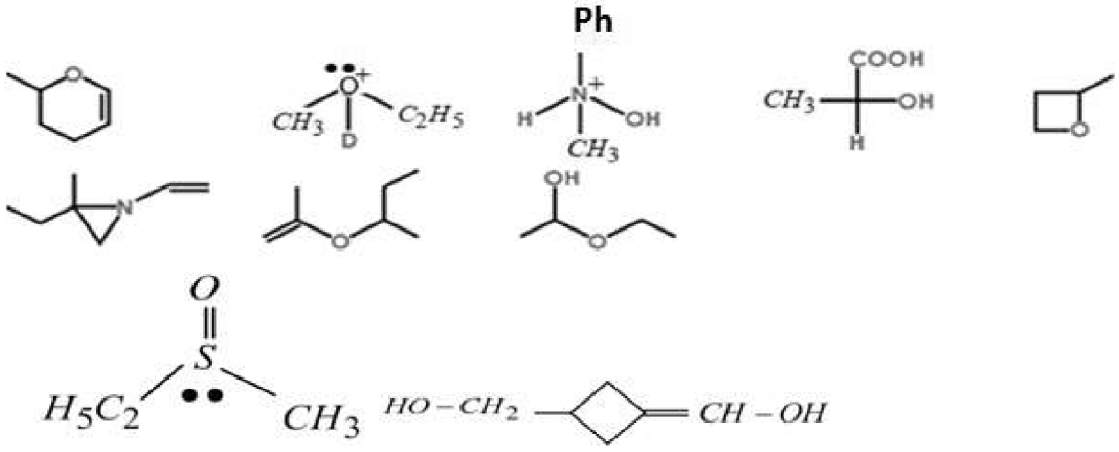
Skipped: –

Q46 Isomerism – Count of optically active compounds

Chemistry · Integer · marking +4 / 0 · marks lost by the top 10: 8 wrong × 4 + 1 skipped × 4 = 36

Lost by 9/10
8 wrong · 1 skipped

46. How many given compounds are optically active



Wrong: Prit Raj Kushwaha, Geetansh Praveen Reddy Vengala, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

Skipped: Ramini Havisha

Q48 **General Organic Chemistry** – Counting electrophiles
Chemistry · Paragraph · marking +3 / 0 · marks lost by the top 10: 8 wrong × 3 = 24

Lost by 8/10
8 wrong · 0 skipped

PARAGRAPH-I

Electrophiles and nucleophiles are fundamental concepts of organic chemistry playing crucial roles in various reactions. Electron loving species that accept electron pairs to form covalent bonds are electrophiles, Nucleus loving species that donate electron pair to form covalent bonds.

48. How many of the following are electrophiles

A) Br^+ B) $\boldsymbol{Na^+}$ C) CN^- D) BF_3

E) $BeCl_2$ F) NH_2^- G) SH^- H) OR^-

Wrong: Prit Raj Kushwaha, Geetansh Praveen Reddy Vengala, Ramini Havisha, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Bandhav Praputh, Pamunugundla Smaran

Skipped: –

Q47 **Hydrocarbons** – Dichlorination isomers of C_5H_{10} (VD 35)
Chemistry · Integer · marking +4 / 0 · marks lost by the top 10: 7 wrong × 4 + 1 skipped × 4 = 32

Lost by 8/10
7 wrong · 1 skipped

47. A hydrocarbon [X] is having vapour density 35 and it forms only 1 monochloro product under sunlight. How many isomeric products (including stereo) are possible when [X] undergoes dichlorination?

Wrong: Prit Raj Kushwaha, Geetansh Praveen Reddy Vengala, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

Skipped: Ramini Havisha

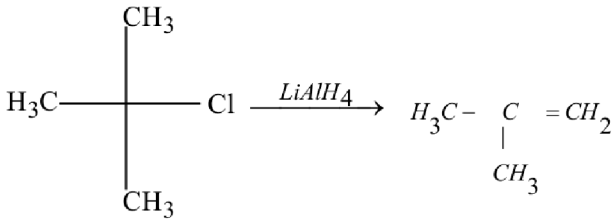
PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

50. Find the No of correct statements out of the following

A) Alkanes are prepared by catalytic hydrogenation of Alkenes where both hydrogen atoms add to the same face of the alkene (syn-addition).

B) Alkyl halides when reacts with Lithium Aluminium Hydride form corresponding alkane: $R-X \xrightarrow{LiAlH_4} R-H$ for R : methyl , $1^0, 2^0$

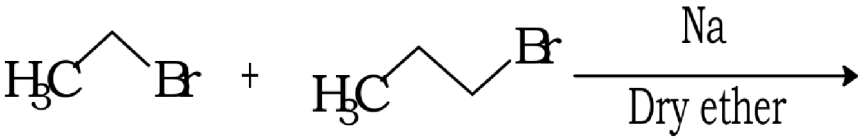


C)In case of Tertiary Alkyl halide is observed.

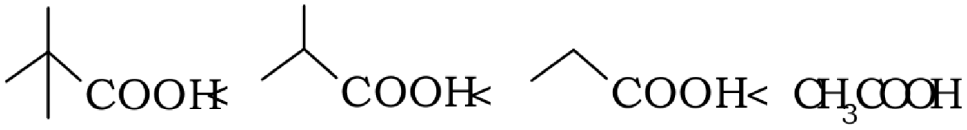
D) Grignard's reagent when treated with a polar protic solvent gives alkane:

$R-Mg+HA \rightarrow R-H+MgX(A)$

E) The different compounds obtained by the following reaction are 7 (only organic compounds)



F) Rate of sodalime decarboxylation for the following acids follow:



G) The side products of potassium butanoate during Kolbe's Electrolysis are propane, propene and propyl butanoate

Wrong: Prit Raj Kushwaha, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Rudroju Rohan Chary, Bandhav Praputh, Pamunugundla Smaran

Skipped: Geetansh Praveen Reddy Vengala

Q43 **Hydrocarbons – Alkane from O_2 needed; count secondary carbons**
Chemistry · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 = 20

Lost by 5/10
5 wrong 0 skipped

43. one mole of alkane (Y) requires 8 moles of oxygen for complete combustion on chlorination with $Cl_2 / h\gamma$, gives four monochlorinated products (including stereo). The total number of secondary carbon atoms per each (Y) is ……..

Wrong: Geetansh Praveen Reddy Vengala, Nandimandalam Loukhya Reddy, Ankam Advaith, Bashaboina Sai Saketh, Bandhav Praputh **Skipped:** –